

Complete Formula Sheet — Numbers Module

(Theory + Exercises + Pre-University Background, with explanations)

Every formula needed to fully work through this module, organized by topic. Each formula now includes a short theoretical explanation.

1. Number Sets — Notation

- $\mathbf{N} = \{1, 2, 3, \dots\}$ — natural numbers. *The counting numbers; the first set humans needed.*
 - $\mathbf{N}^* = \{0, 1, 2, 3, \dots\}$ — naturals including zero. *Same set, with zero added by convention when "nothing" needs to be counted too.*
 - $\mathbf{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ — integers. *Extends $\mathbf{N}$ so that subtraction always has a solution (e.g., $x+2=0$ needs -2).*
 - $\mathbf{Q} = \{a/b : a, b \in \mathbf{Z}, b \neq 0\}$ — rationals. *Extends $\mathbf{Z}$ so that division always has a solution (fractions/parts of a whole).*
 - $\mathbf{I}$ — irrational numbers. *Numbers whose decimal expansion never ends and never repeats — they cannot be written as a fraction (e.g., $\sqrt{2}, \pi$).*
 - $\mathbf{R} = \mathbf{Q} \cup \mathbf{I}$ — real numbers. *The union of every rational and irrational number: the full continuous number line.*
 - $\mathbf{C}$ — complex numbers. *Extends $\mathbf{R}$ so that even equations like $x^2 = -1$ have a solution.*
 - Inclusion chain: $\mathbf{N} \subset \mathbf{Z} \subset \mathbf{Q} \subset \mathbf{R} \subset \mathbf{C}$. *Each set contains the previous one entirely — every natural number is also rational, real, and complex.*
 - " $x \in A$ " = " x belongs to A "; " $A \subset B$ " = " A is a subset of B "; " $\cup$ " = union. *Standard set-theory notation for stating membership and inclusion precisely.*
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2. Basic Algebra (pre-university)

- **Distributive property:** $a(b+c) = ab + ac$. *Multiplying a sum is the same as multiplying each term separately and adding the results — the basis for all polynomial expansion.*
- **Binomial expansion:** $(a+b)(c+d) = ac + ad + bc + bd$. *Applying the distributive property twice: every term of the first factor multiplies every term of the second.*
- **Difference of squares:** $(a+b)(a-b) = a^2 - b^2$. *A special expansion where the "cross terms" ($+ab$ and $-ab$) cancel out — this is exactly why multiplying by a conjugate eliminates the middle term.*
- **Common denominator (fraction addition):** $a/b + c/d = (ad+bc)/bd$. *Fractions can only be added once they represent parts of the same-size whole; converting both to the same denominator makes that possible.*
- **Solving a linear equation:** $ax + b = 0 \rightarrow x = -b/a$. *Isolating the unknown by moving constants to one side and dividing by its coefficient — the simplest kind of equation, solvable whenever $a \neq 0$.*

- **Factoring out a common term:** $xy + xz = x(y+z)$. The reverse of distribution: identifying a factor shared by every term and pulling it outside parentheses, often needed to simplify an expression into a recognizable target form.
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3. Exponents and Radicals (pre-university)

- **Product of powers:** $a^m \cdot a^n = a^{m+n}$. Multiplying the same base repeated m times and then n times is the same as repeating it $m+n$ times in total.
 - **Power of a power:** $(a^m)^n = a^{mn}$. Raising an already-repeated product to another power multiplies the two exponents, since you repeat the " m times" block n times.
 - **Quotient of powers:** $a^m / a^n = a^{m-n}$. Dividing cancels out matching factors, leaving the difference of the exponents.
 - **Product under a root:** $\sqrt[n]{a \cdot b} = \sqrt[n]{a} \cdot \sqrt[n]{b}$ ($a, b \geq 0$). A root distributes over a product, which is what allows simplifying a radical by factoring out a perfect square (or cube, etc.) from underneath it.
 - **Definition of n -th root:** $y = \sqrt[n]{x}$ means $y^n = x$. An n -th root undoes raising to the n -th power — it asks "what number, multiplied by itself n times, gives x ?"
 - **Rationalizing a denominator:** multiply numerator and denominator by the same value (often a conjugate) to remove a radical or imaginary unit from the denominator: $1/\sqrt{2} = \sqrt{2}/2$. Multiplying by (x/x) doesn't change the value of a fraction, but a well-chosen x can turn an "ugly" denominator into a clean rational number.
 - **Factorial:** $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$. The product of all positive integers up to n — this is what generates the coefficients in the infinite series for e^x , $\sin(x)$, and $\cos(x)$.
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4. Quadratic Equations (pre-university)

- **General quadratic equation:** $ax^2 + bx + c = 0$ ($a \neq 0$). Any equation where the unknown appears squared but to no higher power; its graph is a parabola, and the solutions are where that parabola crosses the x -axis.
 - **Quadratic formula:** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ Derived by "completing the square" on the general quadratic; it gives both solutions directly from the coefficients, without needing to factor by inspection.
 - **Discriminant:** $\Delta = b^2 - 4ac$ The expression under the square root in the quadratic formula. Its sign alone tells you what kind of solutions to expect, before even finishing the calculation:
 - $\Delta > 0 \rightarrow$ two distinct real solutions (the parabola crosses the x -axis twice)
 - $\Delta = 0 \rightarrow$ one repeated real solution (the parabola just touches the x -axis)
 - $\Delta < 0 \rightarrow$ two complex conjugate solutions (the parabola never reaches the x -axis, so real solutions don't exist — complex ones fill the gap, using $\sqrt{-1} = i$)
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5. Trigonometry (pre-university)

- **Right-triangle ratios (SOH-CAH-TOA):**
 - $\sin(\theta) = \text{opposite} / \text{hypotenuse}$
 - $\cos(\theta) = \text{adjacent} / \text{hypotenuse}$
 - $\tan(\theta) = \text{opposite} / \text{adjacent} = \sin(\theta)/\cos(\theta)$ *These ratios are constant for a given angle regardless of the triangle's size, which is what lets you relate an angle to a ratio of side lengths (and vice versa).*
- **Pythagorean theorem:** $h^2 = a^2 + b^2$ (h = hypotenuse, a, b = legs). *In any right triangle, the hypotenuse's square equals the sum of the other two sides' squares — this is the geometric fact behind measuring straight-line distance from two perpendicular components.*
- **Radians $\leftrightarrow$ degrees:** $\text{degrees} = \text{radians} \cdot (180/\pi)$; $180^\circ = \pi \text{ rad}$; $360^\circ = 2\pi \text{ rad}$. *Both are just different units for measuring the same rotation — a full circle is 360° or, equivalently, an arc length of 2π times the radius, hence 2π "radians."*
- **Standard angle values** (Table 3 in the module):

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$\sin(\theta)$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos(\theta)$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0

These values come from simple right triangles (the 45-45-90 and 30-60-90 triangles) and are exact, so they're worth memorizing to avoid decimal approximations.

- **Reference-angle identities** (2nd, 3rd, 4th quadrants):
 - 2nd quadrant: $\sin(\theta) = \sin(\pi - \theta)$; $\cos(\theta) = -\cos(\pi - \theta)$
 - 3rd quadrant: $\sin(\theta) = -\sin(\pi + \theta)$; $\cos(\theta) = -\cos(\pi + \theta)$
 - 4th quadrant: $\sin(\theta) = -\sin(-\theta)$; $\cos(\theta) = \cos(-\theta)$ *The unit circle is symmetric, so any angle outside the first quadrant has the same sine/cosine magnitude as some first-quadrant "reference angle" — only the sign changes, depending on which quadrant you're in.*
- **Argument via arctan** (standard $\rightarrow$ polar), with $\theta \in (-\pi, \pi]$:
 - $\theta = \arctan(b/a)$ if $a > 0$
 - $\theta = \arctan(b/a) + \pi$ if $a < 0, b > 0$
 - $\theta = \arctan(b/a) - \pi$ if $a < 0, b < 0$ *Since $\tan(\theta)$ alone doesn't reveal which quadrant the point is in ($\tan$ repeats every π radians), the sign of a and b must be checked separately, and π is added or subtracted to land in the correct quadrant.*
- **Quadrant correction rule of thumb:** a calculator's arctan only returns values in $(-\pi/2, \pi/2)$; if the point actually lies in another quadrant, add or subtract π to get the true

angle. This is the practical version of the rule above: always sketch the point first to know which quadrant to correct toward.

6. Geometry (pre-university)

- **Distance from origin (2D):** $r = \sqrt{x^2 + y^2}$. A direct application of the Pythagorean theorem: the segment from the origin to (x, y) is the hypotenuse of a right triangle with legs x and y .
 - **Vector addition (parallelogram law):** $(a, b) + (c, d) = (a + c, b + d)$. Adding vectors component-by-component is equivalent to placing them tail-to-tail and taking the diagonal of the parallelogram they form — the same operation, seen algebraically and geometrically.
 - **Reflection across the x-axis:** $(a, b) \rightarrow (a, -b)$. Flipping a point vertically over the horizontal axis keeps its horizontal position but inverts its vertical one.
 - **Reflection through the origin:** $(a, b) \rightarrow (-a, -b)$. A 180° rotation about the origin, which flips both coordinates' signs.
 - **Central angle of a regular n -gon:** $360^\circ/n$ (or $2\pi/n$ rad) between consecutive vertices. A regular polygon's vertices are evenly spaced around a circle, so dividing the full turn (360°) by the number of sides gives the angle between any two consecutive vertices.
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7. Sequences, Series & Summation

- **Sigma notation:** $\Sigma(k=1 \text{ to } n) f(k) = f(1) + f(2) + \dots + f(n)$. A compact way to write a long sum that follows a clear pattern, instead of writing out every term with "...".
 - **Sum of first n natural numbers (Gauss):** $\Sigma(k=1 \text{ to } n) k = 1 + 2 + \dots + n = n(n+1)/2$
Discovered by pairing the first and last terms ($1+n$), the second and second-to-last ($2+(n-1)$), etc. — each pair sums to $n+1$, and there are $n/2$ such pairs.
 - **Sum of first n odd numbers:** $1 + 3 + 5 + \dots + (2n-1) = n^2$ A classic result provable by induction; geometrically, it also corresponds to building successive square arrays of dots.
 - **Sum of first n cubes:** $1^3 + 2^3 + \dots + n^3 = [n(n+1)/2]^2$ A remarkable identity: the sum of the first n cubes equals the square of the sum of the first n natural numbers.
 - **Telescoping sum found in the exercises:** $\Sigma(k=1 \text{ to } n) 1/[k(k+1)] = n/(n+1)$ Each term $1/[k(k+1)]$ can be split into $1/k - 1/(k+1)$; when summed, consecutive terms cancel ("telescope"), leaving only the first and last pieces.
 - **n -th odd number formula:** the n -th odd number is $2n-1$. Odd numbers are every other integer starting at 1, so the n -th one is always one less than $2n$.
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8. Natural Numbers: Principle of Induction (module theory)

- **Well-ordering principle:** every non-empty subset of $\mathbb{N}$ has a smallest element. This structural property of the naturals guarantees there's always a "first" case to start from — without it, induction's base case wouldn't make sense.

- **Principle of induction:** let $P(n)$ be a property on N . If:

1. **Base case:** $P(1)$ holds.
2. **Induction step:** for any n , $P(n) \Rightarrow P(n+1)$.

Then $P(n)$ holds for every $n \in N$. Like a row of dominoes: proving the first one falls, and that any falling domino knocks over the next, guarantees they all fall — without ever checking each one individually.

9. Complex Numbers — Standard (Binomial) Form

- **Definition:** $z = a + bi$, where $i^2 = -1$
 - $a = \text{Re}(z)$ (real part), $b = \text{Im}(z)$ (imaginary part) i is defined precisely so that $x^2 = -1$ has a solution; every complex number is then just a real "shift" (a) plus a multiple of this new unit (bi).
 - **Addition/subtraction:** $(a+bi) \pm (c+di) = (a \pm c) + (b \pm d)i$. Since i behaves like an independent unit (similar to a variable in a polynomial), real and imaginary parts combine separately, just like combining "like terms."
 - **Opposite:** $-z = -a - bi$. The additive inverse — the number that, added to z , gives zero.
 - **Multiplication:** $(a+bi)(c+di) = (ac-bd) + (ad+bc)i$. Expand like any two binomials (distributive property), then use $i^2 = -1$ to simplify the resulting i^2 term into a real number.
 - **Conjugate:** $\bar{z} = a - bi$ (if $z = a+bi$). The "mirror" number with the imaginary part's sign flipped — designed specifically to cancel the imaginary part when multiplied by z .
 - **Double conjugation:** $(\bar{z}) = z$. Reflecting twice across the same axis returns you to the original point.
 - **Product of a number and its conjugate (two equivalent forms):**
 - $z \cdot \bar{z} = a^2 + b^2$ (real, ≥ 0)
 - $z \cdot \bar{z} = \text{Re}(z)^2 + \text{Im}(z)^2$ Multiplying $(a+bi)(a-bi)$ is a difference-of-squares pattern: $a^2 - (bi)^2 = a^2 - b^2(-1) = a^2 + b^2$ — the imaginary part vanishes entirely, always leaving a non-negative real number.
 - **Division:** multiply numerator and denominator by the conjugate of the denominator: $z_1/z_2 = (z_1 \cdot \bar{z}_2) / (z_2 \cdot \bar{z}_2)$ Since $z_2 \cdot \bar{z}_2$ is guaranteed to be a real number, this trick turns an "illegal" division by a complex number into an ordinary division by a real one.
 - **Reciprocal shortcuts:**
 - $1/i = -i$
 - $1/(a+bi) = (a-bi)/(a^2+b^2)$ Direct applications of the division rule above, worth memorizing since they appear so often.
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10. Complex Numbers — Polar Form

- **Definition:** $z = r_{\theta}$, where r = modulus ($|z|$), θ = argument. An alternative "address" for the same point: instead of horizontal/vertical distance (a, b), use straight-line distance and direction (r, θ) — like radar coordinates instead of a street grid.
 - **Modulus:** $r = \sqrt{a^2 + b^2}$. The distance from the point to the origin — literally the Pythagorean theorem applied to the real and imaginary parts.
 - **Argument:** $\theta = \arctan(b/a)$, adjusted by quadrant (see trigonometry section). The angle the line from the origin to the point makes with the positive real axis.
 - **Standard $\leftrightarrow$ Polar:**
 - $a = r \cdot \cos(\theta)$, $b = r \cdot \sin(\theta)$
 - $z = r(\cos \theta + i \sin \theta) \leftarrow$ **trigonometric form** This follows directly from the right-triangle definitions of sine and cosine, with the hypotenuse equal to r .
 - **Non-uniqueness of polar form:** $r_{\theta} = r_{(\theta + 2\pi k)}$, for any integer k . Adding a full turn (2π) to the angle points in exactly the same direction, so the same complex number has infinitely many valid angle representations.
 - **Multiplication:** $r_{\theta_1} \cdot s_{\theta_2} = (r \cdot s)_{(\theta_1 + \theta_2)}$. Multiplying moduli scales the distance from the origin, while adding arguments rotates — this becomes clear once complex numbers are written in exponential form (see section 11) and the exponent rule $a^m \cdot a^n = a^{m+n}$ is applied.
 - **Division:** $r_{\theta_1} / s_{\theta_2} = (r/s)_{(\theta_1 - \theta_2)}$. The reverse operation: dividing moduli and subtracting the rotation.
 - **Addition/subtraction:** not direct — convert to standard form first. Polar form has no simple geometric shortcut for addition (unlike multiplication), because adding requires combining horizontal and vertical components separately, which standard form already does.
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11. Complex Numbers — Exponential Form

- **Euler's formula (pure imaginary exponent):** $e^{(\theta i)} = \cos(\theta) + i \cdot \sin(\theta)$. Obtained by substituting $x = \theta i$ into the infinite series for e^x and separating the real terms (which form the cosine series) from the imaginary terms (which form the sine series) — a genuinely surprising bridge between exponential and trigonometric functions.
- **Exponential form:** $z = r \cdot e^{(\theta i)}$ (equivalent to r_{θ}). Simply rewrites the trigonometric form $r(\cos \theta + i \sin \theta)$ using Euler's formula, giving an even more compact notation.
- **General complex exponential (real + imaginary exponent):** $e^{(x + yi)} = e^x \cdot e^{(yi)} = e^x(\cos y + i \sin y)$ Applies the ordinary exponent rule $a^{m+n} = a^m \cdot a^n$ to split off the real exponent (an ordinary real number e^x) from the imaginary exponent (handled by Euler's formula).
- **Product:** $r_1 e^{(\theta_1 i)} \cdot r_2 e^{(\theta_2 i)} = r_1 r_2 \cdot e^{((\theta_1 + \theta_2) i)}$. A direct consequence of the exponent rule $a^m \cdot a^n = a^{m+n}$ applied to base e .

- **Division:** $(r_1 e^{i\theta_1}) / (r_2 e^{i\theta_2}) = (r_1/r_2) \cdot e^{i(\theta_1-\theta_2)}$. Likewise, the exponent rule $a^m/a^n = a^{m-n}$ applied to base e .
 - **Conjugate:** conjugate of $r \cdot e^{i\theta} = r \cdot e^{-i\theta}$. Reflecting across the x -axis flips the sign of the angle while leaving the distance from the origin unchanged.
 - **Powers (De Moivre):** $(r \cdot e^{i\theta})^n = r^n \cdot e^{in\theta}$. An application of $(a^m)^n = a^{mn}$: raising to the n -th power multiplies the modulus by itself n times and multiplies the angle by n (n rotations combined).
 - **Angle reduction for large powers:** if the resulting angle $n\theta = 2\pi m + \phi$ for integer m , then $e^{in\theta} = e^{i\phi}$. Since adding 2π doesn't change the direction (non-uniqueness of the angle, section 10), any whole number of extra turns can simply be dropped to simplify the angle.
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12. Roots of Complex Numbers (Roots of Unity)

- **Number of solutions:** equation $z^n = w$ always has exactly n solutions in $\mathbb{C}$. Unlike in $\mathbb{R}$ (where, e.g., negative numbers have no real square root), every nonzero complex number has exactly n distinct n -th roots — a consequence of the non-uniqueness of the angle (any of infinitely many equivalent angles can be divided by n to give a genuinely different result).
 - **General n -th root formula:** given $z^n = r \cdot e^{i\theta}$, the n solutions are: $z = \sqrt[n]{r} \cdot e^{i(\theta/n + 2\pi k/n)}$, $k = 0, 1, 2, \dots, n-1$. Since z 's original angle could equally be written as $\theta + 2\pi k$ for any integer k , dividing by n before taking the root produces n genuinely different angles (for $k=0$ to $n-1$) before the pattern repeats.
 - **Geometric fact:** the n roots are equally spaced on a circle of radius $\sqrt[n]{r}$, forming the vertices of a regular n -gon, separated by an angle of $2\pi/n$. Each increase of k by 1 adds exactly $2\pi/n$ to the angle while keeping the modulus fixed — exactly the condition for evenly-spaced points on a circle (section 6).
 - **Cube roots of unity ($z^3=1$):** $1, e^{i(2\pi/3)}, e^{i(4\pi/3)} = 1, -1/2 + \sqrt{3}/2 i, -1/2 - \sqrt{3}/2 i$. The specific case $n=3, r=1, \theta=0$ of the general root formula above — three points forming an equilateral triangle on the unit circle.
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13. Series Expansions Behind Euler's Formula (background theory)

- **Exponential series:** $e^x = 1 + x + x^2/2! + x^3/3! + x^4/4! + \dots$. An infinite sum that defines e^x for any x ; adding more terms gives a better and better approximation, and factorials in the denominators make later terms shrink rapidly.
 - **Sine series:** $\sin(x) = x - x^3/3! + x^5/5! - x^7/7! + \dots$. The infinite-series definition of sine, built only from odd powers of x with alternating signs.
 - **Cosine series:** $\cos(x) = 1 - x^2/2! + x^4/4! - x^6/6! + \dots$. The infinite-series definition of cosine, built only from even powers of x with alternating signs. (Substituting $x = \theta i$ into the exponential series and separating real/imaginary terms produces exactly the sine and cosine series above — this is where Euler's formula comes from.)
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Master Quick-Reference Table

#	Formula	Topic
1	$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$	Number sets
2	$(a+b)(c+d) = ac+ad+bc+bd$	Algebra
3	$(a+b)(a-b) = a^2-b^2$	Algebra
4	$a^m \cdot a^n = a^{m+n}, (a^m)^n = a^{mn}$	Exponents
5	$\sqrt{(ab)} = \sqrt{a} \cdot \sqrt{b}$	Radicals
6	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	Quadratics
7	$\Delta = b^2 - 4ac$	Quadratics
8	$\sin = \text{opp/hyp}, \cos = \text{adj/hyp}, \tan = \text{opp/adj}$	Trig
9	$h^2 = a^2 + b^2$	Pythagoras
10	$\text{rad} \leftrightarrow \text{deg}: 180^\circ = \pi \text{ rad}$	Trig
11	Table of $\sin/\cos$ at $0, \pi/6, \pi/4, \pi/3, \pi/2$	Trig
12	$\theta = \arctan(b/a) \pm \text{quadrant correction}$	Trig
13	$(a,b) + (c,d) = (a+c, b+d)$	Vectors
14	$\sum k \ (k=1..n) = n(n+1)/2$	Series
15	$1+3+\dots+(2n-1) = n^2$	Series
16	$1^3+\dots+n^3 = [n(n+1)/2]^2$	Series
17	$\sum 1/[k(k+1)] = n/(n+1)$	Series
18	$P(1) \wedge [P(n) \Rightarrow P(n+1)] \Rightarrow \forall n P(n)$	Induction
19	$i^2 = -1$	Complex — definition
20	$(a+bi) \pm (c+di) = (a \pm c) + (b \pm d)i$	Complex — standard
21	$(a+bi)(c+di) = (ac-bd) + (ad+bc)i$	Complex — standard
22	$z \cdot \bar{z} = a^2 + b^2$	Complex — standard
23	$1/i = -i; 1/(a+bi) = (a-bi)/(a^2+b^2)$	Complex — standard

#	Formula	Topic
24	$z=r_{\theta}; r=\sqrt{a^2+b^2}$	Complex — polar
25	$r_{\theta}=r_{(\theta+2\pi k)}$	Complex — polar
26	$r_{\theta_1} \cdot s_{\theta_2}=(rs)(\theta_1+\theta_2); r_{\theta_1}/s_{\theta_2}=(r/s)(\theta_1-\theta_2)$	Complex — polar
27	$e^{i\theta}=\cos \theta + i \sin \theta$	Complex — Euler
28	$e^{(x+yi)}=e^x(\cos y + i \sin y)$	Complex — exponential
29	$(r \cdot e^{i\theta})^n=r^n \cdot e^{in\theta}$	Complex — De Moivre
30	$z=\sqrt[n]{r} \cdot e^{i((\theta/n+2\pi k/n))}, k=0..n-1$	Complex — roots
31	$e^x, \sin x, \cos x$ power series	Background for Euler's formula